Respiration and Circulation

Assignment and notes

1. **Respiration** is a biochemical process of oxidation of organic compounds in an orderly manner for the liberation of chemical energy in the form of ATP.
2. The nasal cavity is divisible into right and left nasal chambers by a **mesethmoid cartilage**.
3. The **nasopharynx** is the uppermost part from the nasal chamber it leads into **oropharynx.**
4. Oropharynx continues below as the **laryngopharynx.**
5. The pharynx has a set of lymphoid organs called **tonsils**.
6. **Larynx** is called voice box.
7. The **larynx** extends from the laryngopharynx and the hyoid bone to the trachea.
8. The larynx opens into the layngopharynx through a slit like opening called glottis. This opening of the trachea or wind pipe is guarded by a leaf like flap called **epiglottis.**
9. **Epiglotis** prevents the entry of food into trachea.
10. The **trachea** divides into right and left primary bronchi as it reaches the middle of the thoracic cavity.
11. The bronchi are supported internally by 'C' shaped incomplete rings of **cartilage**.
12. Each lobe of the lung has the terminal bronchioles ending in a bunch of air sacs, each with **10 to 12** alveoli.
13. **Diaphragm**is a muscular septum that separates the thoracic and abdominal cavity.
14. **Tidal volume (T.V)**is the volume of air inspired or expired during normal breathing. It is 500 ml.
15. **Inspiratory reserve volume (IRV) is** The maximum volume of air, or the extra volume of air, that is inspired during forced breathing in addition to T.V. Its value is 2000 to 3000ml.
16. **Expiratory reserve volume (ERV) is** The maximum volume of air that is expired during forced breathing after normal expiration. Its value is 1000 to 1100ml.
17. **Dead space (DS) is** The volume of air that is present in the respiratory tract (from nose to the terminal bronchioles), but not involved in gaseous exchange. It is 150 ml.
18. **Residual volume (RV) is** The volume of air that remains in the lungs and the dead space even after maximum expiration. It is 1100 to 1200ml.
19. **Total Lung capacity is** The maximum amount of air that the lungs can hold after a maximum forcefull inspiration (5200 to 5800ml).
20. **Vital capacity (VC) is** The maximum amount of air that can be breathed out after a maximum inspiration. It is the some total of TV, IRV and ERV and is 4100 to 4600ml.
21. **Haemoglobin** acts as the respiratory carrier.
22. Theoretically, one molecule of Hb has **4 Fe^{++},** each of which can pick up a molecule of **oxygen (O_2).**
23. Hence Haemoglobin can carry **8** oxygen atoms i.e**4** O_2 molecules.
24. the degree of saturation of Hb with O_2 depends upon the **O_2 tension i.e. ppO_2.**
25. Maximum saturation of 95 to 97% is at ppO_2 in alveoli (100 mmHg).
26. At 30 mmHg of ppO_2, only 50% saturation can be maintained.
27. The relationship between HbO_2 saturation and oxygen tension (ppO_2) is called **oxygen dissociation curve**
28. **Bohr effect :**It is the shift of oxyhaemoglobin dissociation curve due to change in partial pressure of CO_2 in blood.
29. **Haldane effect :**Oxyhaemoglobin functions as an acid. It decreases pH of blood. Due to increase in the number of H^+ ions, HCO^-_3 changes into H_2O and CO_2.

30. To maintain the ionic balance between the RBCs and the plasma, Cl^- diffuses into the RBCs. This movement of chloride ions is known as **chloride shift** or **Hamburger's phenomenon**
31. In the RBCs, CO_2 combines with water in the presence of a Zn containing enzyme, **carbonic anhydrase** to form carbonic acid.
32. At the **level of the lungs** in response to the low partial pressure of carbon dioxide ($ppCO_2$) of the alveolar air, hydrogen ionand bicarbonate ions recombine to form carbonic acid and under the influence of carbonic anhydrase yields carbon dioxide and water.
33. Carbon dioxide binds with the amino group of the haemoglobin and form a loosely bound compound carbaminohaemoglobin.
34. ATP is used to carry out vital life processes and so is called as energy currency of the cell.
35. Steady rate of respiration is controlled by neurons located in the **pons** and **medulla** and are known as the **respiratory centres**
36. The inspiratory muscles relax and expiration follows. As air leaves the lungs during expiration, the lungs are deflated and the stretch receptors are no longer stimulated. Thus, the inspiratory centre is no longer inhibited and a new respiration begins. These events are called the ***Hering-Breuer reflex***.
37. The ***Hering-Breuer reflex*** controls the depth and rhythm of respiration.
38. **Artificial ventilation** is also called artificial respiration.
39. A ventilator is a machine that supports breathing and is used during surgery, treatment for serious lung diseases or other conditions when normal breathing fails.
40. In round worms there are no blood vessels and the body fluid is moved around the viscera by contraction of body wall and muscles. This is **extracellular transport**.
41. **Cyclosis**is the streaming movement of the cytoplasm shown by almost all living organisms e.g. *Paramoecium*, *Amoeba*, root hair cells of many plants and WBCs in animals.
42. It is for transportation within the cell or **intracellular transport**.
43. **Open circulation**e.g. Arthropods and Molluscs
44. **Single circulation** e.g. fishes
45. **Double circulation e.g.** birds and mammals
46. animals, heart pumps deoxygenated blood to lungs for oxygenation and it returns to heart as oxygenated blood. This is '**pulmonary circulation**'.
47. The oxygenated blood is pumped from the heart towards various body parts (except lungs) and returns back to the heart as deoxygenated blood. This is '**systemic circulation**'.
48. Human heart shows <u>**double**</u> circulation.
49. The process of formation of RBCs is called **erythropoiesis**.
50. Vitamin B_{12}, folic acid and heme protein are required for production of RBCs.
51. Condition with increase in the number of RBCs is called **polycythemia**
52. **D**ecrease in number of RBCs is called as **erythrocytopenia**.
53. The hormone <u>**erythropoietin**</u> produced by the kideny cells stimulates the bone marrow for production of RBCs
54. The **hematocrit** is ratio of the volume of RBCs to total blood volume of blood.
55. Leucocytes are colourless, nucleated and amoeboid cells larger than RBCs.Due to their amoeboid movement they can move out of the capillary walls by a process called **diapedesis**.
56. Decrease in number of WBCs (<4000) is called **leucopenia**
57. Temporary increase in number of WBCs is called as **leucocytosis**
58. Uncontrolled increase in number of WBCs is a type of blood cancer called **leukemia**.
59. Neutrophils are responsible for destroying pathogens by the process of **phagocytosis**.
60. 'Pus' is mixture of dead neutrophils, damaged tissues and dead microbes.
61. <u>**Eosinophils**</u> are also responsible for detoxification as they produce antitoxins.

62. If number of thrombocytes decreases than normal, condition is called as **thrombocytopenia**. This condition causes internal bleeding (haemorrhage).
63. Platelets secrete platelet factors which are essential in blood clotting.
64. Platelets also seal the ruptured blood vessels by formation of **platelet plug/ thrombus**.
65. Platelets secrete**serotonin** a local **vasoconstrictor**.
66. Heart is enclosed in a membranous sac called **pericardium**.
67. Pericardium is formed of two main layers - outer **fibrous** and inner **serous pericardium.**
68. Heart is conical in shape and lies in **mediastenum-** i.e. the space between two lungs.
69. The human heart is four chambered. The two superior chambers are called **atria** (auricles) and inferior two are called **ventricles**.
70. Externally, the atria are separated from ventricles by a transverse groove called **coronary sulcus** or **atrioventricular groove**.
71. The two ventricles are externally separated from each other by two grooves, the anterior and posterior **inter-ventricular sulci**.
72. The right atrium recieves**superior** and **inferior vena cava** along its dorsal surface.
73. Inter auricular septum has an oval depression called **fossa ovalis**. It is a remnant of the embryonic aperture called **foramen ovalis**.
74. Superior vena cava (precaval), inferior vena cava (postcaval) and coronary sinus open into the right atrium. Opening of the postcaval is guarded by a **Eustachian valve** while the **Thebesian valve** guards the opening of coronary sinus into right atrium.
75. Both the atria open into the ventricles of their respective sides by atrioventricular apertures. These openings are guarded by cuspid valves. The **tricuspid valve** is present in the right AV aperture and **bicuspid valve** (mitral valve) is present in the left AV aperture.
76. Contraction of heart muscles is **systole** and relaxation of heart muscles is **diastole.**
77. A single systole followed by diastole makes one **heart beat.**
78. The heart beats 70 to 72 times per minute. This is called **heart rate**.
79. During each heart beat ventricles pump about 70 ml of blood this is called **stroke volume**.
80. It means heart pumps about 72 (heart rate) x 70 ml (stroke volume) = 5040 ml ≈ 5 liters of blood per minute this is called **cardiac output**.
81. The human heart is **myogenic** i.e. the heart is capable of generating a cardiac contraction independent of nervous input.
82. A part of this nodal tisse is present in the upper right corner of the right atrium. It is called **SA Node** or **Sino-atrial node**.
83. Another mass of nodal tissue, the modified muscular fibers also called autorhythmic fibers (conducting tissue)**<u>AV node</u>** control the beating rate of heart.
84. Consecutive systole and diastole constitutes a single heartbeat or **cardiac cycle**.
85. Cardiac cycle is completed in **0.8 sec**.
86. On an average, 72 beats are completed in one minute in an adult, at rest.
87. **Bundle of His/ Tawara**branches start from AV node and pass through interventricular septum.
88. In normal conditions, atrial systole is for 0.1 sec. and atrial diastole (AD) is for 0.7 sec.
89. For a normal adult human being it is calculated as follows : (CO) = SV × HR = 70 × 72 = 5040 ml/min.
90. When all the chambers of heart are in diastole, this condition is called **joint diastole** or **complete diastole**. Thus, duration of one cardiac cycle is 0.8 sec
91. Chemical control of the heart rate includes the conditions like hypoxia, acidosis, alkalosis causing decreased cardiac activity, hormones like **<u>epinephrine and norepinephrine</u>** enhance the cardiac activity
92. Cardiac activity **<u>decreases</u>** with the elevated blood level of K_+ and Na_+

93. Difference between systolic and diastolic pressure is called **<u>pulse pressure</u>**. Normally, it is 40 mmHg.

94. Deviations from normal blood pressure value indicate malfunctioning of heart. It may be due to high or low blood volume, arterial inelasticity or hardening of arteries(**<u>arteriosclerosis</u>**), deposition of fats like cholesterol in the arteries (a**<u>therosclerosis</u>**), renal diseases and emotion induced hormonal changes, obesity, etc.

95. Blood pressure is measured with the help of an instrument called **<u>sphygmomanometer</u>**.

96. An optimal blood presure (normal) level reads **120/80 mmHg**.

97. Persistently raised blood pressure higher than the normal is called hypertension

98. **Coronary Artery Disease (CAD) :-** It is also known as atherosclerosis

99. **Angina Pectoris :**It is the pain in the chest resulting from a reduction in the blood supply to the cardiac muscles because of atherosclerosis or arteriosclerosis.

100. **Angiography :** X-ray imaging of the cardiac blood vessels to locate the position of blockages is called angiography

101. Graphical recording of electrical variations detected at the surface of body during their propagation through the wall of heart is **electrocardiogram (ECG)**.

102. **P-wave** represents the atrial depolarization.

103. The **QRS complex**. QRS complex represents the ventricular depolarization.

104. **T-wave** It represents the ventricular repolarization..

105. Inflammation of tonsils is called **<u>as tonsillitis.</u>**

106. Viral tonsillitis cures naturally but bacterial tonsillitis needs antibiotic treatment.

107. **<u>Tonsillectomy</u>** is performed in some patients who do not respond to the antibiotic treatment.

<u>Notes</u>

Q. 1 Choose the correct alternatives from those given below and complete the statements.

1. The muscular structure that separates the thoracic and abdominal cavity is _______.

a. pleura b. **<u>diaphragm</u>** c. trachea d. epithelium

2. What is the minimum number of plasma membrane that oxygen has to diffuse across to pass from air in the alveolus to haemoglobin inside a R.B.C.?

a. **Tw**o b. Three c. Four d. Five

3. ________ is a sound producing organ,

a. **<u>Larynx</u>** b. Pharynx c. Tonsils d. Trachea

4. The maximum volume of gas that is inhale during breathing in addition to T.V is _____.

a. residual volume b**<u>. I.R.V</u>**. c. G.R.V. d. vital capacity

5. ________ muscles contract when the external intercostal muscles contract

a. Internal abdominal b. Jaw c. Muscles in bronchial walls d. **<u>Diaphragm</u>**

6. Movement of cytoplasm in unicellular organisms is called __________.

a. diffusion b. **<u>cyclosis</u>**c. circulation. d. thrombosis.

7. Which of the following animals do not have closed circulation?

a. **<u>Earthworm</u>**. b. Rabbit c. Butterfly d. Shark

8. Diapedesis is performed by _________.

a. erythrocytes b. thrombocytes c. adipocytes d**<u>. leucocytes</u>**

9. Pacemaker of heart is _________.

a. **<u>SA node</u>** b. AV node c. His bundle d. Purkinje fibers

10. Which of the following is without nucleus?

a. **<u>Red blood corpuscle</u>** b. Neutrophill c. Basophill d. Lymphocyte

11. Cockroach shows which kind of circulatory system?

a. Open b. **<u>Closed</u>** c. Lymphatic d. Double

12. Diapedesis can be seen in _________ cell.

a. RBC b. **WBC** c. Platelet d. neuron

13. Opening of inferior vena cava is guarded by __________.

a. bicuspid valve b. tricuspid valve c. Eustachian valve d. **Thebesian valve**

14. ___________ wave in ECG represent atrial depolarization.

a. **P** b. QRS complex c. Q d. T

15. The fluid seen in the intercellular spaces in Human is _________

a. blood b. lymph c. **interstitial fluid** d. water

Q. 2 Match the Respiratory surface to the organism in which it is found

Respiratory surface	**Organism**
Plasma membrane	Insect
Lungs	Salamander
External gills	Bird
Internal Gills	Amoeba
Trachea	Fish

Q. 3 Very short answer questions.

1. Why does trachea have 'C' shaped rings of cartilage?

Ans: The cartilaginous rings are C-shaped to allow the trachea to collapse slightly at the opening so that food can pass down the esophagus

2. Why is respiration in insect called direct respiration?

Ans:Insects have Open circulation system : In animals having an open circulation, blood is circulated through the body cavities (haemocoels). Exchange of material takes place directly between blood and cells or tissues of the body.Hence it is called direct respiration.

3. Why is gas exchange very rapid at alveolar level.

Ans: The partial pressure of carbon dioxide in the capillaries is much higher than that in the alveoli. This means that net diffusion occurs into the alveoli from capillaries. The carbon dioxide can then be exhaled as the partial pressure in the alveoli is also higher than the partial pressure in the external environment.

4. Name the organ which prevents the following the entry of food into the trachea while eating.

Ans: The epiglottis is a leaf-shaped flap in the throat that prevents food from entering the windpipe and the lungs.

Q 4.Short answer questions.

1. **Why is it advantageous to breathe through the nose than through the mouth?**

   Ans:

   i. We can breathe through our nose and mouth.Breathing through nose is better than breathing through mouth. Because,The hairs inside nostrils filter any microbe in the air and it only lets the clean air pass through.

   ii. The air gets warm and humidified as it passes into our bodies, making it less shocking to our systems.

   iii. Each nostril takes part in filtering, warming, moisturizing, dehumidifying, and smelling the air.

   iv. Lungs can extract oxygen from the air we breathe primarily on the exhale.

   v. Because the nostrils are smaller than the mouth, air exhaled through the nose creates back pressure when one exhales.

   vi. Work of lung movement would get doubled.

vii. We can breathe through mouth in emergencies, vigorous activities like, work out, sports etc.

2. Identify the incorrect statement and correct it,

a. A respiratory surface area should have a large surface area.: Correct statement

b. A respiratory surface area should be kept dry: incorrect statement

Ans: Oxygen from the air dissolves momentarily in the **surface** water, allowing more time for the oxygen to cross the membrane of the alveoli. ... **If** the **surface dries out**, the exchange of gas will take place at a very low pace as quickly moving gaseous oxygen molecules do not cross the alveoli membrane effectively

c. A respiratory surface area should be thin, may be 1mm or less.: correct statement

3. Given below are the characteristics of some modified respiratory movement. Identify them.

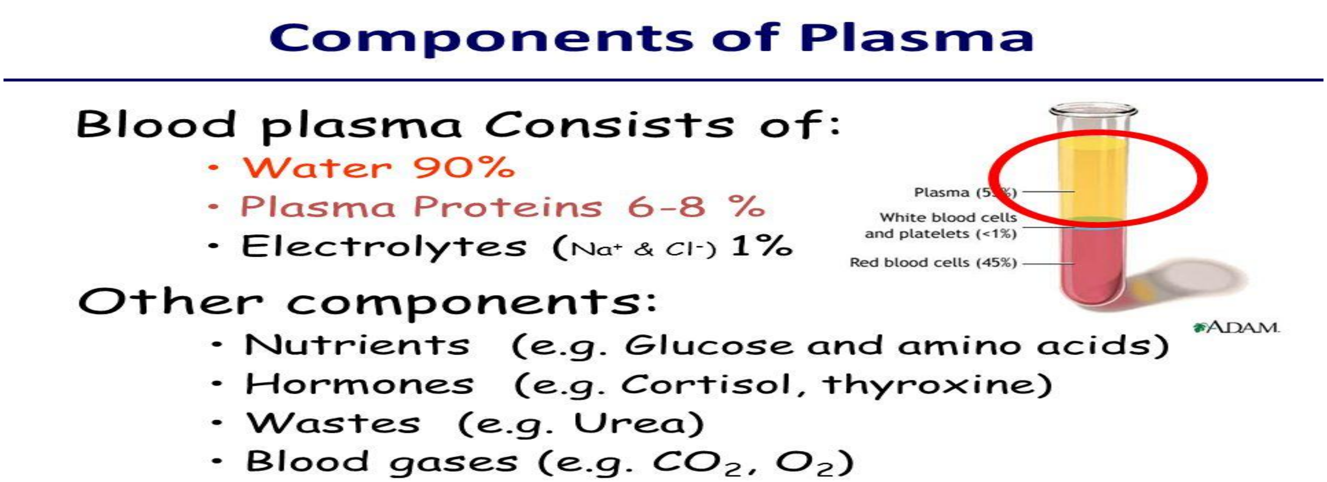
a. Spasmodic contraction of muscles of expiration and forceful expulsion of air through nose and mouth.

Ans; Sneezing

b. An inspiration followed by many short convulsive expiration accompanied by facial expression.

Ans: Coughing

4. Write a note on blood plasma.



<u>**Composition of Blood Plasma**</u>

1. Water :90%
2. Proteins (albumen, globulin, properdin, prothrombin, fibrinogen):6-8%
3. Inorganic salts (Na, K, Mg, Ca, Fe, Mn and Cl^-, HCO_3^- and PO_4^{3-}) :1%
4. Others :

a. Food (glucose, amino acids, fatty acids, triglycerides)

b. Wastes (urea, uric acid and creatinine)

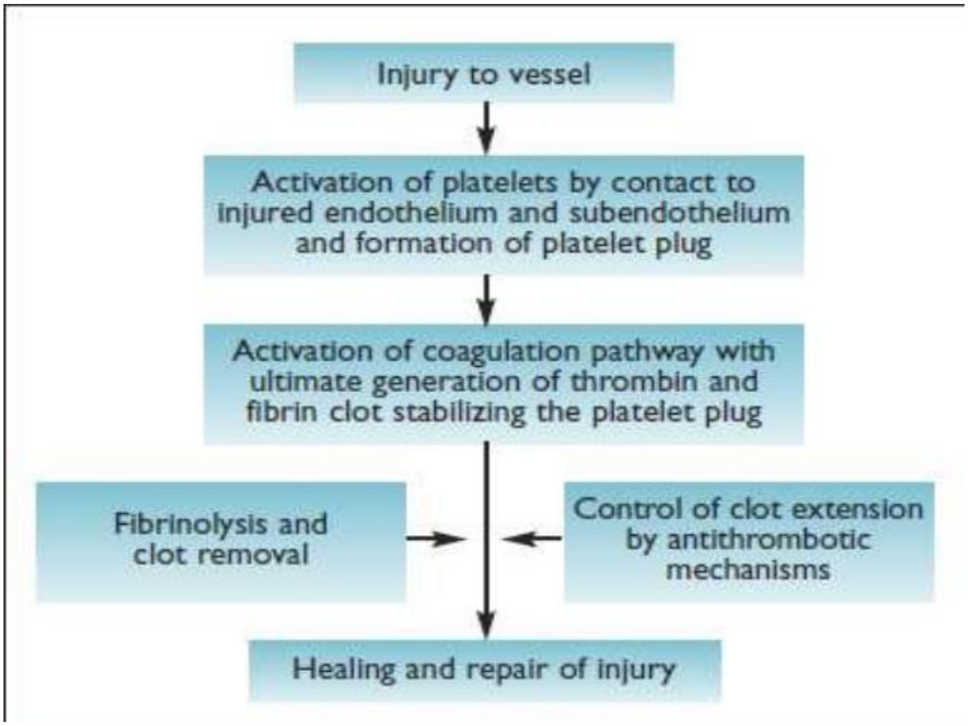
c. Regulators (hormones, enzymes, vitamins)

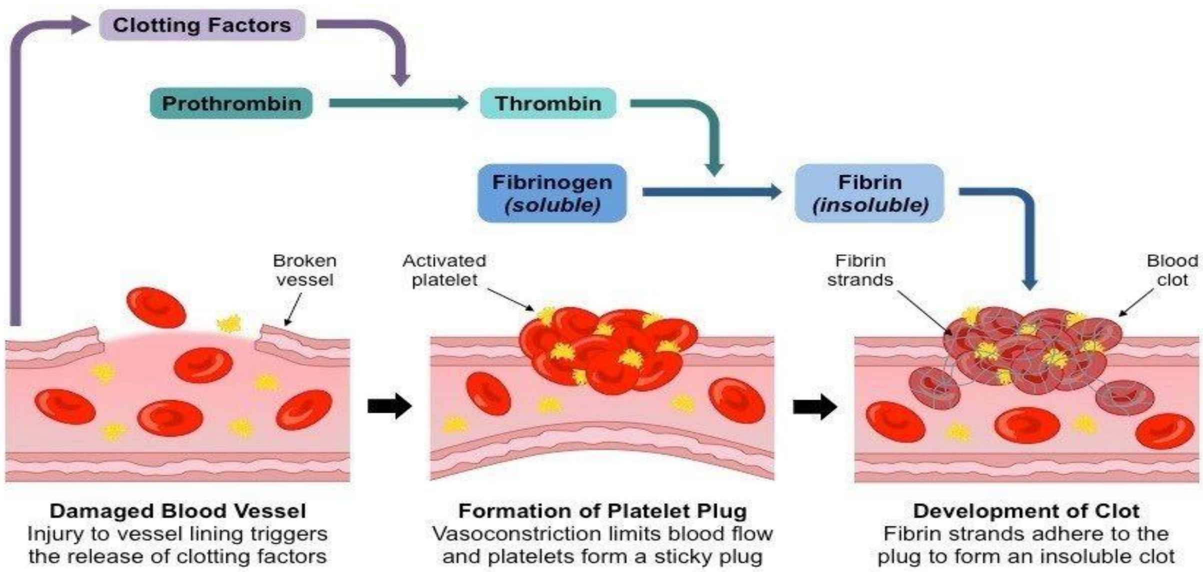
d. Anticoagulants (heparin)

e. Cholesterol and antibodies

f. Dissolved gases (O_2, CO_2, N_2)

5. Explain blood clotting in short.





6. Describe pericardium.

Ans: The **pericardium** is the fibrous sac that surrounds the heart. ... The parietal and visceral pericardia together form the serous **pericardium**. The two layers of the serous **pericardium** are continuous with each other. The fibrous **pericardium** is a layer of connective tissue that provides support and protection for the heart.

7. Describe valves of human heart.

Ans:

1. The heart has 4 chambers, 2 upper chambers (atria) and 2 lower chambers (ventricles).
2. Blood passes through a valve before leaving each chamber of the heart.

3. The valves prevent the backward flow of blood.
4. Valves are actually flaps (leaflets) that act as one-way inlets for blood coming into a ventricle and one-way outlets for blood leaving a ventricle.
5. Normal valves have 3 flaps (leaflets), except the mitral valve.
6. It only has 2 flaps. The 4 heart valves are:

   a)Tricuspid valve. This valve is located between the right atrium and the right ventricle.

b) Pulmonary valve. The pulmonary valve is located between the right ventricle and the pulmonary artery.

c) Mitral valve. This valve is located between the left atrium and the left ventricle. It has only 2 leaflets.

d) Aortic valve. The aortic valve is located between the left ventricle and the aorta.

7. **What is role of papillary muscles and chordae tendinae in human heart?**
   Ans: The **papillary muscles** are **muscles** located in the ventricles of the **heart**. They attach to the cusps of the atrioventricular valves (also known as the mitral and tricuspid valves) via the **chordae tendineae** and contract to prevent inversion or prolapse of these valves on systole (or ventricular contraction).
8. **Explain in brief the factors affecting blood pressure.**
   Ans: 1. The three factors that contribute to blood pressure are resistance, blood **viscosity**, and blood vessel diameter.
2. **Increase in blood pressure:**
   Blood pressure increases with increased cardiac output, peripheral vascular resistance, volume **of blood**, viscosity **of blood** and rigidity **of** vessel walls.
3. **Decrease in blood pressure:**
   Blood pressure decreases with decreased cardiac output, peripheral vascular resistance, volume **of blood**, viscosity **of blood** and elasticity **of** vessel walls.

Q. 5 Give scientific reason.

1. Closed circulation is more efficient than open circulation.

Ans: The blood is pumped through a **closed system** of arteries, veins, and capillaries. ... Still, blood can flow backward and the **system** is only slightly **more efficient** than the **open system** of insects. The **closed circulatory system** has **more** advantages over the **open circulatory system**

2. Human heart is called as myogenic and autorhythmic.

Ans: In the human **heart**, contraction is initiated by a specially modified **heart** muscle known as the sinoatrial node. ... Since the heartbeat is initiated by the SA node and the impulse of contraction originates in the **heart** itself, the human

heart is termed as **myogenic**. The **hearts** of vertebrates and mollusks are also **myogenic**

4. **Person who has undergone heart transplant needs lifetime supply of immunosupressants.**
   Ans: After an organ **transplant**, you will need to **take immunosuppressant** (anti-rejection) **drugs**. These **drugs** help prevent your immune system from attacking ("rejecting") the donor organ. Typically, they must be **taken** for the lifetime of your **transplanted** organ
5. **Arteries are thicker than veins.**
   Ans: Arteries experience a pressure wave as blood is pumped from the heart. This can be felt as a "pulse." Because of this pressure the walls of **arteries** are much **thicker than** those of **veins**. ... The vessel walls of **veins** are thinner **than arteries** and do not have as much tunica media.
6. **Left ventricle is thick than all other chambers of heart.**
   Ans: Arteries experience a pressure wave as blood is pumped from the heart. This can be felt as a "pulse." Because of this pressure the walls of **arteries** are much **thicker than** those of **veins**. ... The vessel walls of **veins** are thinner **than arteries** and do not have as much tunica media.

Q. 6 Distinguish between :

1. Open and closed circulation.

CARDIOVASCULAR SYSTEM
VERSUS
CIRCULATORY SYSTEM

CARDIOVASCULAR SYSTEM	CIRCULATORY SYSTEM
Cardiovascular system is the organ system through which the blood circulates	Circulatory system is the system by which blood and lymph circulate in the body
Composed of the heart and blood vessels	Composed of the heart, blood vessels, blood, lymphatic vessels, glands, and lymph
An anatomic term	A physiologic term
Consists of blood vessels	Consists of blood and lymph
A closed system	Lymphatic system is an open system
Provide oxygen and nutrients to the body cells while removing the metabolic wastes	Also has a defensive role in addition to the functions of cardiovascular system

2. Artery and vein.

Artery	Vein
Carries blood away from heart	Carries blood toward heart
Blood under high pressure	Blood under low pressure
Thick walls	Thin walls
Pulse flow	Smooth flow
Narrow lumen	Large lumen
No valves	Valves present
Blood rich in oxygen (except pulmonary artery)	Blood poor in oxygen (except pulmonary veins)

3. Blood and lymph.

A.

Blood	Lymph
1) Reddish in colour.	1) Pale yellow in colour.
2) Red blood cells are present.	2) Red blood cells are absent.
3) Bidirectional flow.	3) Unidirectional flow.
4) Flow is rapid.	4) Flow is slow.
5) Leucocyte count relatively less.	5) High leucocyte count.
6) Platelets present.	6) Platelets absent.

4. Blood capillary and lymph capillary.

S. No.	Lymph Capillaries	Blood Capillaries
1.	Colourless, difficult to observe	Reddish, easy to observe
2.	Blind (closed at the tip)	Joined to the arterioles at one to venules at the another end
3.	Wider than blood capillaries	Narrow than lymph capillaries
4.	Contain colourless lymph	Contain red blood

5. Intrinsic and extrinsic process of clotting.

COMPARISON BETWEEN INTRINSIC AND EXTRINSIC MECHANISM OF BLOOD CLOTTING

FEATURES	INTRINSIC MECHANISM	EXTRINSIC MECHANISM
SOURCE OF THROMBOPLASTIN	Damaged platelets, Intrinsic source	Damaged tissue, Extrinsic source
ROLE OF FACTOR VII	Not required	Required
NUMBER OF REACTIONS	More	Less
TIME	Takes few minutes	Few seconds
STARTS FROM ACTIVATION OF	Factor XII	Factor VII

Q. 7 Long answer questions.

1. Smita was working in a garage with the doors closed and automobiles engine running. After some time she felt breathless and fainted. What would be the reason? How can she be treated?

Ans:Reason:

The extremely high concentrations of carbon monoxide produced by an **engine can** raise CO concentrations in a **closed** building so quickly that a person **may** collapse before **they** even realize there is a problem. Carbon monoxide reduces the amount of oxygen **to** the brain, causing CO poisoning, and lack of reasoning.

Treatment: The best way to **treat CO poisoning** is to breathe in pure oxygen. This **treatment** increases oxygen levels in the blood and helps to remove **CO** from the blood. Your doctor will place an oxygen mask over your nose and mouth and ask you to inhale.

2. Shreyas went to a garden on a wintry morning. When he came back, he found it difficult to breath and started wheezing. What could be the possible condition and how can he be treated?

Ans;1. The **cold** temperatures **can** trigger symptoms such as wheezing, coughing, and **shortness of breath**. Even in healthy people, **cold**, dry air **can** irritate the airways and lungs. It **causes** the upper airways to narrow, which makes it a little harder to **breathe.**

2. Expiratory wheezing alone often **indicates** a mild airway obstruction. ... Inspiratory **wheezing** often accompanies **expiratory wheezing** when heard over the lungs, specifically in acute asthma. However, if inspiratory **wheezing** or stridor is heard over the neck, that could be an indication of a serious upper airway obstruction

1. **3.Treatment:** Drink warm liquids. ...
2. Inhale moist air. ...
3. Eat more fruits and vegetables. ...
4. Quit smoking. ...
5. Try pursed lip breathing. ...
6. Don't exercise in cold, dry weather

3. Why can you feel a pulse when you keep a finger on the wrist or neck but not when you keep them on a vein?

Ans:

4. A man's pulse rate is 68 and cardiac output is 5500 cm^3.Find the stroke volume.

Ans: Cardiac output= 5,500 liters/min

Heart rate =68 bpm

We can use the equation CARDIAC OUTPUT = HR x SV.

5500= 68 X SV

Therefore SV= 5500/68

SV = 80.882 ML

STROKE VOLUME = 80.882 ML

5. Which blood vessel of the heart will have the maximum content of Oxygen and why?

ANS: The pulmonary artery channels oxygen-poor blood from the right ventricle into the lungs, where oxygen enters the bloodstream. The **pulmonary veins** bring oxygen-rich blood to the left atrium. The aorta channels oxygen-rich blood to the body from the left ventricle.

Hence Pulmonary veins have the maximum content of Oxygen.

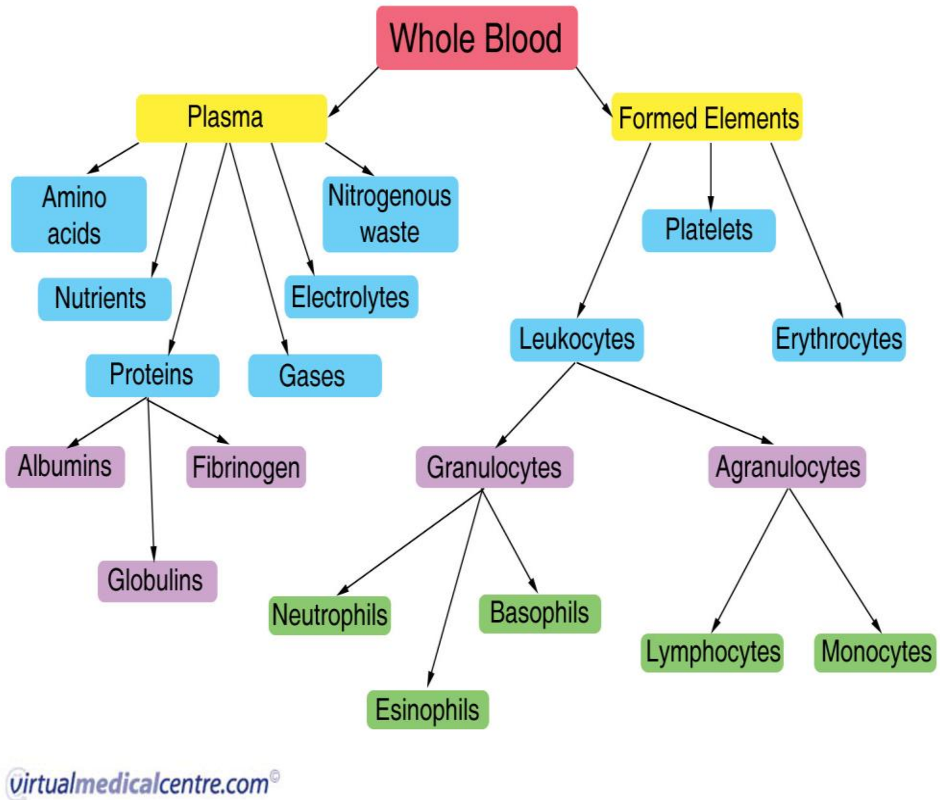
7. **If the duration of the atrial systole is 0.1 sec and that of complete diastole is 0.4 sec, then how does one cardiac cycle complete in 0.8 sec?**

   Ans: The atrial systole takes 0.1 seconds. And the atrial diastole takes 0.7 seconds {the total is 0.8 seconds = the cardiac cycle}. We have AV delay; so that the atria and ventricles do not contract at the same time. Therefore, atrial systole is followed by ventricular systole.

8. **How is blood kept moving in the large veins of the legs?**

   Ans: The answer is the muscles of your body working with the one-way valves. As the muscles of your body contract and relax to move your limbs they also squeeze your veins. ... Because of the one-way valves, the blood can only keep moving toward the heart

10. Describe human blood and give its functions.



Erythrocytes are circular biconcave non-nucleated cells.
2. In males there are about 5.1 to 5.8 million RBC's per cubic mm while in females 4.3 to 5.2 million RBC's per cubic mm
3. The average life span of RBC is 120 days.
4. Formation of RBC's is called **erythropoiesis.**
5. Increase in number of RBC's is called **polychthemia.**
6. Decrease in number of RBC's is called **erythrocytopaenia.**
7. Cytoplasm of RBC's contain respiratory pigment called **haemoglobin** which transports oxygen &CO_2 .
8. Normal haemoglobin in adult is 13-18 gm/100 ml of blood while in female 11.5 to 16.5 gm/100 ml of blood.
9. Less amount of haemoglobin leads **anemia.**
10. **Functions of RBC's**: *It transports O_2 from lungs to tissue while CO_2 from tissue to lungs. They maintain blood pH as haemoglobin acts as buffer. They also maintain viscosity of blood.*

i) Formation of WBC's is called leucopoesis
ii) It occurs in red bone marrow, spllen lymph nodes, tonsils, thymus and Payer's patches.
iii) Increase in number of WBC's may cause leucocytosis
iv) Decrease in its number is called leucopaenia
v) Leukaemia is excessive increase in number of WBC's

2.The three major types of white blood cells are:

- Granulocytes
- Monocytes
- Lymphocytes

Granulocytes

i) There are three different forms of granulocytes:

- Neutrophils
- Eosinophils
- Basophils

ii) Granulocytes are phagocytes, that is they are able to ingest foreign cells such as bacteria, viruses and other parasites.

iii) Granulocytes are so called because these cells have granules of enzymes which help to digest the invading microbes. Granulocytes account for about 60% of our white blood cells.

iv) Neutrophils are by far the most prevalent of these cells. Each neutrophil cell can ingest up to between around 5 and 20 bacteria in its lifetime.

v) Eosinophils are involved in allergic reactions and can attack multicellular parasites such as worms.

vi) Basophils are also involved in allergic reactions and are able to release histamine, which helps to trigger inflammation, and heparin, which prevents blood from clotting.

vii) Agranulocytes

It consists of monocytes and lymphocytes

Monocytes

i) Monocytes can develop into two types of cell:

- **Dendritic cells** are antigen-presenting cells which are able to mark out cells that are antigens (foreign bodies) that need to be destroyed by lymphocytes.
- **Macrophages** are phagocyte cells which are larger and live longer than neutrophils. Macrophages are also able to act as antigen-presenting cells.

Lymphocytes

vi) Lymphocytes are cells which help to regulate the body's immune system.

vii) The main types of lymphocytes are:

a) B lymphocytes (B cells) b) T lymphocytes (T cells)